
CE 505: Engineering of Ground Modification

Course Code	CE 505
Course Name	Engineering of Ground Modification
L-T-P-C	3-0-0-3
Prerequisites	CE 302 - Geotechnical Engineering
Intended for	UG/PG
Distribution	Discipline Core
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to Engineering Ground Modification**
Need and objectives, Identification of soil types, In-situ and laboratory tests to characterize problematic soils; Mechanical, Hydraulic, Physico-chemical, Electrical, Thermal methods and their applications. [8 Lectures]
- **Module II**
Mechanical Modification – Principles of soil densification – Properties of Compacted soil, Compaction control tests, Specification of compaction requirements, Blasting Vibrocompaction, Dynamic Tamping and Compaction piles. [8 Lectures]
Hydraulic Modification – Objectives and techniques, traditional dewatering methods and their choice, Design of dewatering system, Electro-osmosis, Filtration, Drainage and seepage control with Geosynthetics, Preloading and vertical drains, Electro-kinetic dewatering. [8 Lectures]
- **Module III**
Physical and Chemical Modification – Modification by admixtures, Shotcreting and Guniting Technology, Modification at depth by grouting, Crack Grouting and compaction grouting, Jet grouting, Thermal Modification, Ground freezing. [9 Lectures]
- **Module IV**
Modification by Inclusions and Confinement - Soil reinforcement, underpinning, In-situ ground reinforcement, ground anchors, rock bolting and soil nailing, sand bags, crib walls, bin walls, gabion walls. [7 Lectures]
- **Module V**
Introduction to geosynthetics – Applications. [3 Lectures]

Text Books

1. Peter G. Nicholson, **Soil Improvement and ground modifications methods**, Elsevier, 2015.
2. P. Purushothama Raj, **Ground improvement Techniques**, Laxmi Publications, 2005.
3. Hausmann, M. R., **Engineering Principles of Ground Modifications**, McGraw Hill publications, 1990.

References

1. John A. Hudson, **Ground Improvement Case Histories**, Elsevier, 2005.
2. C. A. Raison, **Ground and Soil Improvement**, ICE publications, 2004.